CIE AS PHYSICS (9702) • QUESTION PAPER

8.2 Diffraction

Syllabus Section: 8. Superposition **Paper Type:** Multiple Choice (Paper 1) **Total:** 15 Questions (15 Marks)

Question 1 Easy [1 mark]

Which of the following is the definition of diffraction?

A change of direction when waves cross the boundary between one medium and another

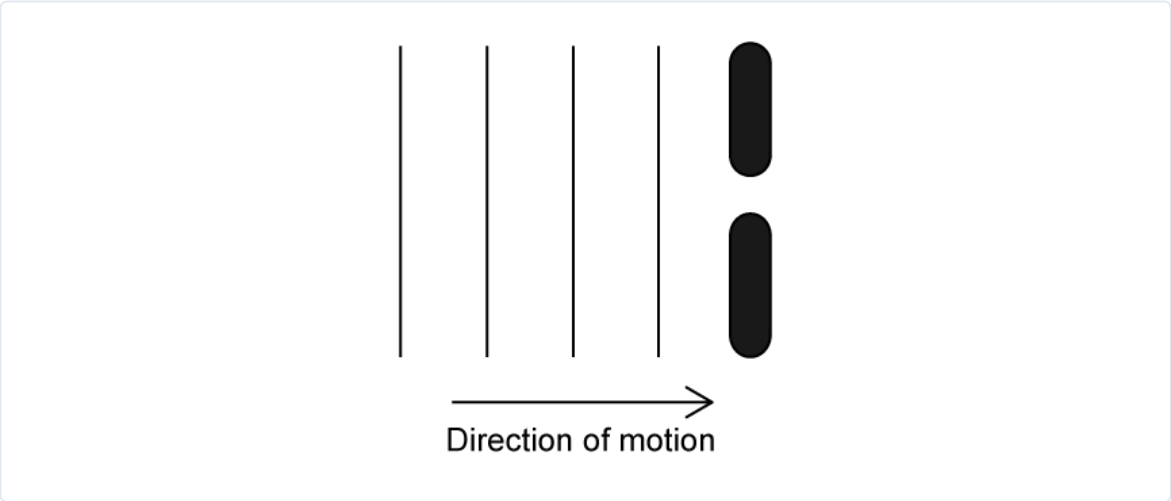
B splitting of white light into colours

C addition of two coherent waves to produce a stationary wave pattern

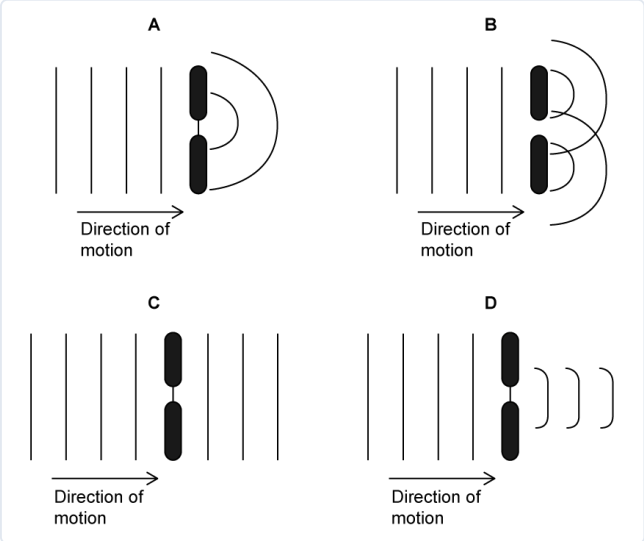
D bending of waves around an obstacle

Question 2 Easy [1 mark]

The diagram shows a wave approaching an aperture.



Which diagram correctly shows the resulting diffraction pattern for the wave?



A

B

C

D

Question 3 Easy [1 mark]

For which types of wave does diffraction occur?

A In all waves

B In soundwaves only

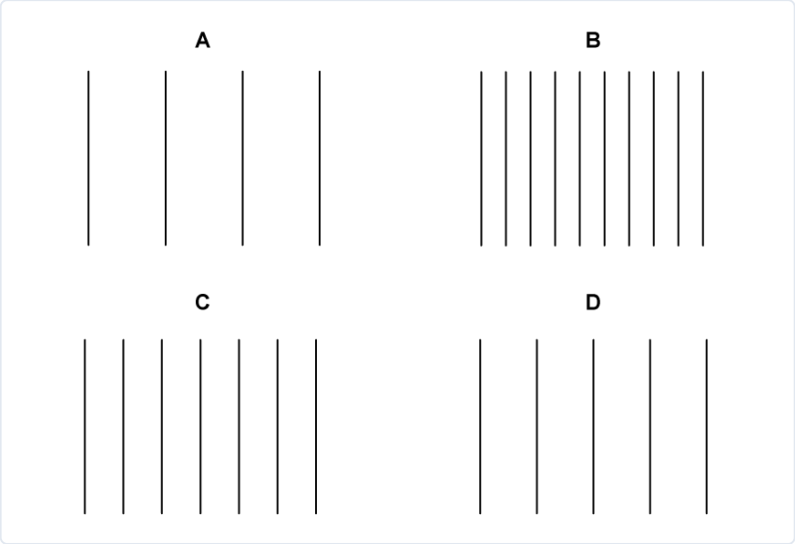
C In transverse waves only

D In light waves only

Question 4 Easy [1 mark]

Wavefront diagrams can be used to indicate the frequency of waves.

Which wave has the lowest frequency?



A

B

C

D

Question 5 Easy [1 mark]

Which answer option gives a correct description of wave diffraction?

A A wave is returned from its original direction when incident on a shiny surface

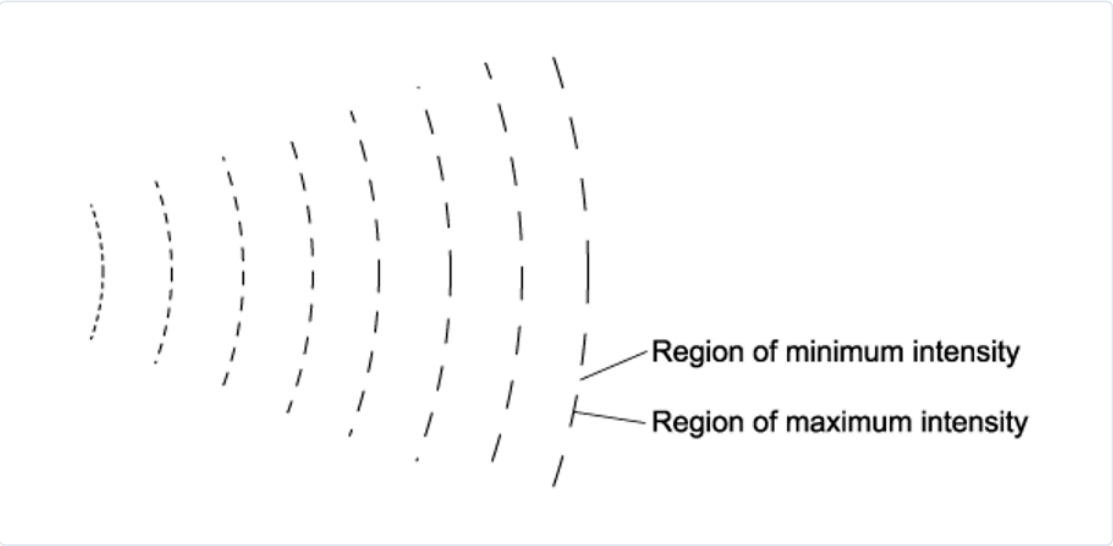
B A wave changes direction when it passes between materials of different density

C Waves spreading out when they pass an obstruction

D The constructive interference of two waves in phase

Question 6 Medium [1 mark]

The diagram shows the pattern of waves. The source of the waves was unable to be seen.



Which of the following would cause this pattern?

A diffraction only

B interference only

C coherence only

D diffraction and interference

Question 7 Medium [1 mark]

An atomic lattice has a spacing of around 10^{-10} m, which electromagnetic wave would cause the most significant diffraction effect?

A X-ray

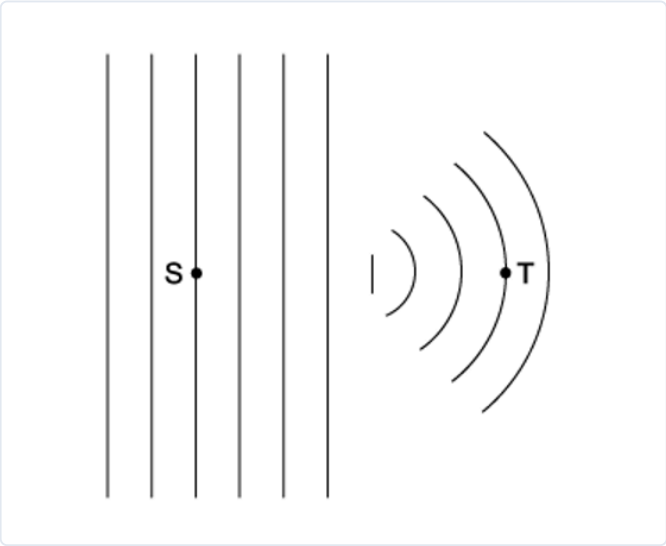
B infra-red

C microwave

D ultraviolet

Question 8 Medium [1 mark]

A ripple tank was used to demonstrate plain wavefronts passed through a gap, as shown in the diagram.



Which property would be different at S compared to T?

A frequency

B amplitude

C wavelength

D velocity

Question 9 Medium [1 mark]

When a wave passes an obstruction diffraction can be observed, this effect is greatest when the wavelength and obstruction are similar in size.

When waves are travelling through the air, which of the following would best demonstrate diffraction?

A radio waves passing a copper wire

B visible light waves passing a gate post

C microwaves passing a steel post

D sound waves passing human hair

Question 10 Medium [1 mark]

A slit of width 0.2 m has sound waves incident upon it. The speed of sound is 340 ms^{-1}.

What does the frequency of the sound wave need to be for the greatest diffraction to occur?

A 500 Hz

B 1000 Hz

C 1700 Hz

D 3000 Hz

Question 11 Hard [1 mark]

Where wavelength is given as λ, and w is the width of the aperture, which of the following answer options would result in the greatest diffraction?

A $\frac{\lambda}{w} \approx 1$

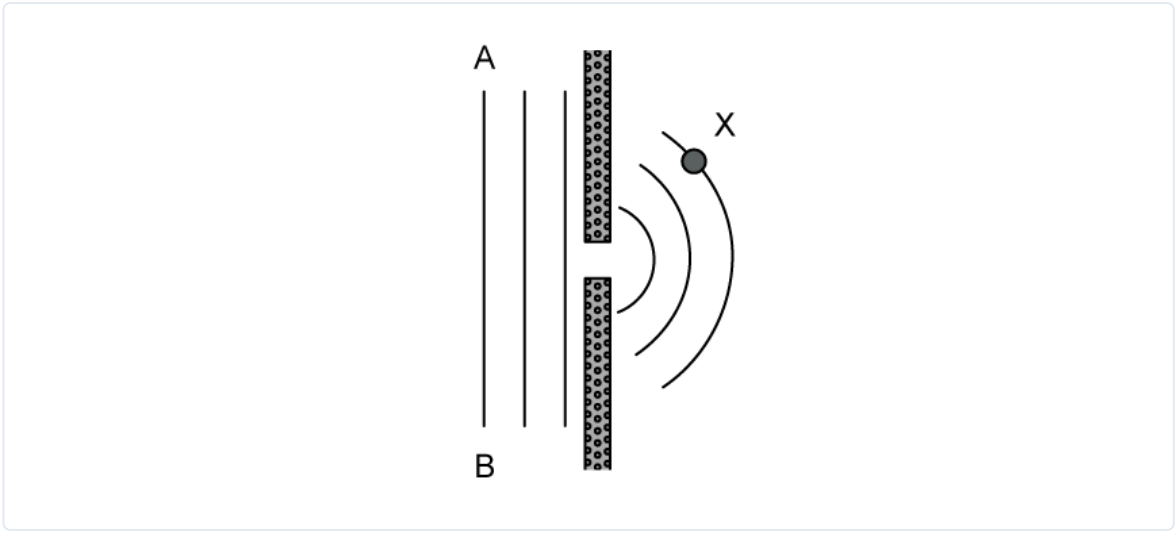
B $\frac{\lambda}{w} < 1$

C $\frac{\lambda}{w} \ll 1$

D $\frac{w}{\lambda} > 1$

Question 12 Hard [1 mark]

The diagram shows wavefronts in a ripple tank diffracting through an aperture. The waves have a speed, *c*, and a wavelength, *λ*.



Which of the following expressions gives the time taken for the wavefront AB to reach point X?

A $\frac{3\lambda}{c}$

B $\frac{6\lambda}{c}$

C $\frac{3\lambda}{3c}$

D $\frac{5\lambda}{c}$

Question 13 Hard [1 mark]

A 5 cm cube casts a shadow when a torch is shone from behind it. When a speaker is placed behind it emitting sound of a frequency 4000 Hz, the sound can still be detected behind the cube.

Which of the following statements explain this observation?

A Light travels faster than sound

B Light is a transverse wave and sound is longitudinal

C Light waves have shorter wavelengths than sound waves

D Light waves are electromagnetic and sound waves are compression waves

Question 14 Hard [1 mark]

Sound diffracts around a loudspeaker aperture of diameter 15 cm.

The speed of sound is 340 m s^{-1}, and the frequency of the sound is 3 kHz.

What effect would be observed?

A No diffraction

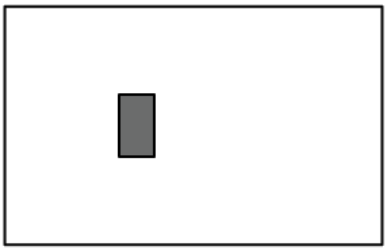
B Slight diffraction

C Significant diffraction

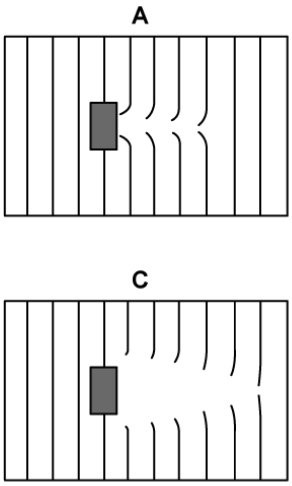
D Not enough information to tell

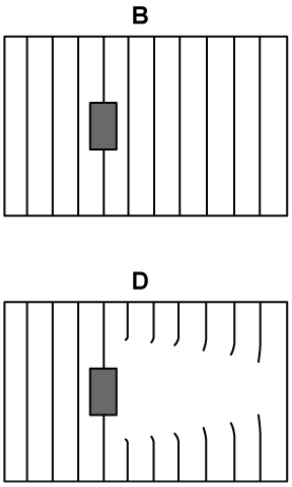
Question 15 Hard [1 mark]

A block is placed in a ripple tank as shown in the diagram.



Which answer option shows the effect on the wavefront pattern observed?





A

B

C

D